

Judge's Commentary: The Outstanding Velociraptor Papers

John S. Robertson
Dept. of Mathematics and Computer Science
Georgia College and State University
Milledgeville, GA 31061-0490
jroberts@mail.gac.peachnet.edu

Introduction

The life sciences provide a particularly fertile area in which a mathematical modeler can apply the craft. Physics and engineering are the traditional playgrounds of applied mathematicians, while biology and its subdisciplines have often been thought of as areas in which little or no mathematics is necessary—soft subjects, if you will. That stereotype is just not true. (Just look at any volume of *The UMAP Journal*!)

Paleontology is a particularly rich area for the mathematical modeler because there are simply no data available from observations. It is not possible, for example, to watch a tyrannosaur hunt down its dinner or a pterodactyl soar through the air. What we do know about ancient animals is based on a lot of inferential detective work, something that often requires a substantial amount of applied mathematics.

The success of Michael Crichton's novels *Jurassic Park* and *The Lost World* (and their extraordinary popularity as movies under the masterful direction of Steven Spielberg) only served to raise the interest in many people—young ones, in particular—in paleontology. In that sense, the timing for this problem could not have been better.

The Importance of Assumptions

The modeling process hinges critically on the assumptions made. In the cases of the velociraptor and the thescelosaurus, paleontologists provided data about certain physical characteristics of these dinosaurs, based on analysis of fossil remains. The paleontologists needed help in analyzing hunting strategies used by the raptors.

The UMAP Journal 18(3)(1997)293–295. ©Copyright 1997 by COMAP, Inc. All rights reserved. Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice. Abstracting with credit is permitted, but copyrights for components of this work owned by others than COMAP must be honored. To copy otherwise, to republish, to post on servers, or to redistribute to lists requires prior permission from COMAP.

The data provided suggested that unrealistically high forces were present as either animal made a sharp turn, a maneuver almost certainly necessary for survival. Most teams realized this; and since they were unable to obtain revised data from the paleontologists, they made reasonable moderating assumptions. This type of difficulty is not unknown in the real world, and it was important to the success of the teams that they identified the data provided as containing an evident weakness and reacted in some reasonable and appropriate way. For example, quite a few teams drew on information readily available in the literature concerning mammalian species similar in size and behavior traits to the dinosaurs in the problem. This enabled the teams to adjust the data in a realistic way, raising the likelihood that their subsequent results would be of real utility to their clients.

Another major issue that had to be treated with assumptions dealt with the geometry and mechanisms of the stalk, the chase, and the capture. The best teams provided clear, detailed thinking about their choices. Good work here inevitably eased the problem of interpretation that the teams eventually had to face.

The Choice of Models

Once teams clarified their assumptions, they set to the task using a surprising diversity of approaches. Some teams were able to formulate models that used just algebra and geometry and no calculus. Others used differential equations, and one of the best papers submitted used differential game theory.

In almost every case, teams turned to a computer to perform model calculations. The judges saw all manner of approaches to this. Computer algebra systems were very popular, as was Matlab. And many teams used a good old-fashioned programming language (C++ in most cases) for this purpose.

This problem was especially well suited to graphical interpretation of one sort or another, and most teams provided graphs and charts that depicted the conduct and outcome of chases with one or two predators. Graphical analysis is particularly important when working with clients who may not grasp all the technical mathematics (such as paleontologists). The old saying—that a picture is worth a thousand words—is decidedly true in this case, as the illustrations produced by teams were absolutely vital to their analysis or the model's predictions.

Analysis of Results

The very best papers gave a thorough treatment to an interpretation of their results and predictions. In most case, this involved a consideration of model weaknesses uncovered by those results. These weaknesses often were traced back to assumptions made at the beginning of the process. This step is also

important to the modeler. The client is not always in a position to assess the existence, never mind the seriousness, of weaknesses. Such weaknesses are not necessarily fatal, but they do serve to point out where better data, more accurate assumptions, or different methods are called for. This leads to an important interaction between the modeler and the client. Without this, the whole process loses its focus.

Conclusion

The Outstanding papers showed remarkable depth of insight into both the biology and the mathematics used, as well as into the process of melding the two together. The judges had a great time reading them. It was evident that the topic stimulated a intense interest on the part of the contest participants. No doubt the dinosaur problem has prompted many students to contemplate investigating biology as an exciting and fruitful area of work for a mathematician.

About the Author

Dr. John S. Robertson is Chair and Professor of Mathematics and Computer Science at Georgia College and State University, the alma mater of the great southern writer Flannery O'Connor. He describes himself as a "dirt-under-the-fingernails" applied mathematician, and is as fascinated by the applications of mathematics to other disciplines as he is by the mathematics itself. He and his family are happily ensconced in the Deep South, from where he enjoys watching weather reports about all the winter snow that falls in the northeastern United States. He doesn't miss it a bit.